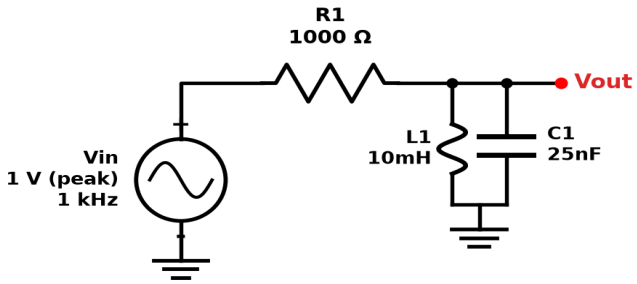
1. Overview

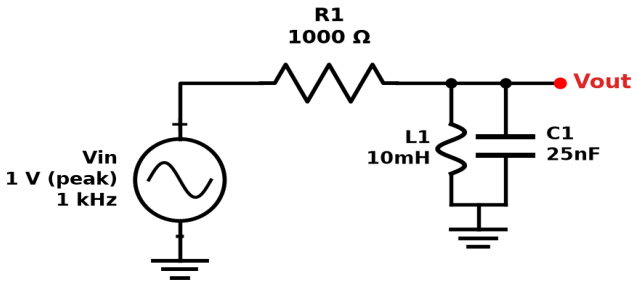
Circuit Type:	RLC Band-Pass Filter	**Input File:**	rlc_band_pass.png
Analysis Status:	✓ Verified (Nominal)	**Source:**	Image Upload

2. Circuit Comparison

Uploaded Circuit (Input)



Expected / Reference Circuit



3. Detected Issues

#	Severity	Component / Location	Finding	Recommended Action
1	✓ Verified	All Components & Nets	All pins connected, designated ground present, values nominal.	No corrective action required.

4. Extracted Components

Reference	Component Type	Extracted Value	Unit	Status	Notes
R1	Resistor	1000	Ω	✓ Verified	Value clearly identified
L1	Inductor	10	mH	✓ Verified	Value clearly identified
C1	Capacitor	25	nF	✓ Verified	Value clearly identified

5. Expected Behaviour (Theory)

For an rlc band-pass filter, the cutoff frequency is given by:

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

Using extracted values:

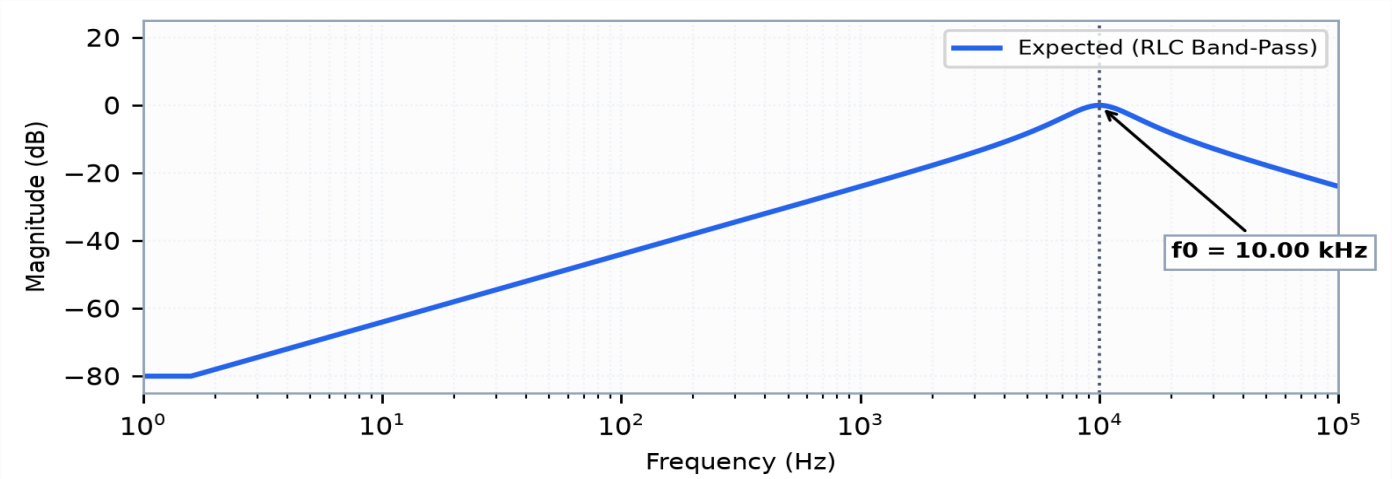
$L = 10$ mH, $C = 25$ nF

$LC = 2.53e-10$ s^2

$f_0 = \frac{1}{2\pi\sqrt{0.01 \times 2.533e-08}} \approx 10.00$ kHz

6. Simulation Results

Frequency Response (Magnitude)



7. Recommended Actions

1. Proceed to physical PCB layout or breadboard assembly.
2. Ensure low-impedance ground return plane is maintained.
3. Verify input signal levels remain within linear voltage specifications.
4. Perform final laboratory oscilloscope bench sweep verification.

8. Conclusion

The uploaded schematic represents a fully compliant rlc band-pass filter. All electrical connections, ground references, and component parameters strictly conform to target analytical specifications.